OBJECT-ORIENTED PROGRAMMING (OOP) MASTER TEXTBOOK

1. Fundamental Concepts

Class vs. Object

A **Class** is a blueprint or template for creating objects. It defines the state (data) and behavior (methods). An **Object** is a specific instance of a class, occupying actual memory.

2. The Four Pillars of OOP

Encapsulation

Bundling data and methods together and restricting direct access to some of an object's components (using private/protected access modifiers).

Abstraction

Hiding complex implementation details and showing only the necessary features of an object.

Inheritance

A mechanism where a new class (subclass) derives properties and behavior from an existing class (superclass).

Polymorphism

The ability of an object to take on many forms. This is achieved via method overloading (compile-time) and method overriding (run-time).

3. Constructors and Destructors

Lifecycle Management

Constructors: Special methods called automatically when an object is created to initialize data members.

Destructors: Methods called when an object is destroyed, used to release memory or system resources.

4. Implementation Example (Java-style)

```
class Car {
private String model; // Encapsulation
public Car(String m) { model = m; } // Constructor
public void drive() { System.out.println("Driving " + model); }
}
// Inheritance example
class ElectricCar extends Car { ... }
```